3.1

a. Standard deviation measures variation between individual outcomes. Standard error measures variation between estimates across possible random assignments.

b. Under the sharp null, treatment affects nobody. We repeat every possible random assignment, calculate the effect each time, and compare these results with the observed effect. The proportion that is equally or more extreme is the p value.

c. A 95 per cent confidence interval gives a range of effects consistent with the data. Across repeated experiments, about 95 per cent of intervals created this way would contain the true ATE.

d. Complete random assignment allocates individuals from the full sample. Block random assignment allocates individuals separately within similar groups. Cluster random assignment allocates whole groups.

e. A balanced design places similar numbers in treatment and control. This usually lowers the standard error and improves precision.

3.5 a

Village 1: - 0.83
Village 2: 0
Village 3: 15.83
Village 4: 0.83
Village 5: 4.17
Village 6: 0
Village 7: 15

3.5 b

35/7 = 5
True ATE: 5
The average of all possible estimates equals the true ATE, so the estimator is unbiased.